

LAB 2A:

[Omega . A Computer Algebra System Explorer]

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f: (h and not c) or (p and not c);
/* Test all combinations of inputs */
f, h=false, p=false, c=false;
f, h=false, p=false, c=true;
f, h=true, p=false, c=false;
f, h=true, p=false, c=true;
f, h=false, p=true, c=false;
f, h=false, p=true, c=true;
f, h=true, p=true, c=false;
f, h=true, p=true, c=true;

```

$$h \wedge \neg c \vee p \wedge \neg c$$

false

false

true

false

true

false

true

false

☒ Algebra
 ☐ Trig
 ☐ Matrix
 ☐ Calculus
 ☐ Other

Execute

x^2	$\sqrt{x}$	e^x	$\log_e$	π	e	i	$-\infty$	∞
$ x $	mod	nPr	nCr	(	)	,	:	;
ev	subst	rat-	rad-	7	8	9	^	=
gcd	lcm	p?	factor	4	5	6	*	/
ex+	pfrac	?r	solve	1	2	3	+	-
				0	.	E	←	CLR
				2D	3D	SP	DEL	↵

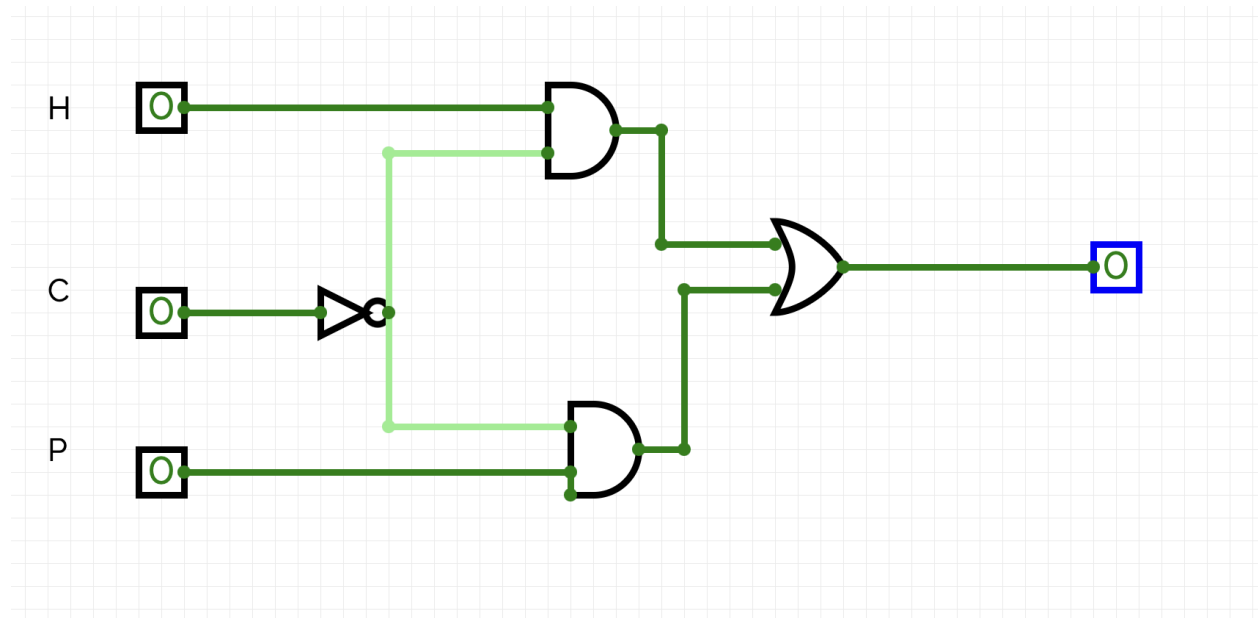
☒ Expert
 ☒ Maxima

In this problem, I've to draw the circuit for an automatic door which should open if a person passes through it, or by pressing a switch, but only if the door is unlocked. This is the behavior I model with the following logic equation for door opening: $f = (h \text{ and not } c) \text{ or } (p \text{ and not } c)$, where f is the door opening, h is the switch holding the door open, p is a person being detected and c is the door being locked. The line "f:“(h and not c) or (p and not c);” specifies the logic equation with the help of Omega in Maxima editor. Then, I plug in several combinations of h , p , and c values to the formula employing such lines as f , $h=false$, $c=true$, $p=false$; and it gives a simulation of whether the door opened or closed depending on the input. This way, I can ensure that the logic equation works just as intended for every single possible value of the net input current. Attached below is a table that goes through all the inputs:

P	H	C	Expected F
0 (False)	0 (False)	0 (False)	0 (False)
0 (False)	0 (False)	1 (True)	0 (False)
0 (False)	1 (True)	0 (False)	1 (True)
0 (False)	1 (True)	1 (True)	0 (False)
1 (True)	0 (False)	0 (False)	1 (True)
1 (True)	0 (False)	1 (True)	0 (False)
1 (True)	1 (True)	0 (False)	1 (True)
1 (True)	1 (True)	1 (True)	0 (False)

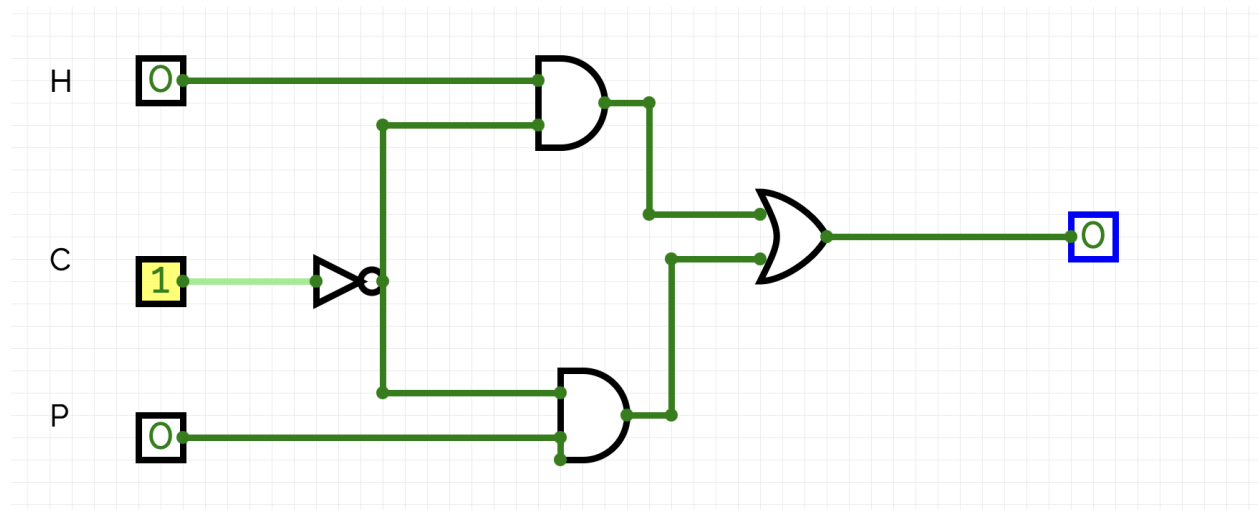
LAB 2B:

When $P = 0$, $H = 0$, $C = 0$, the expected is 0:



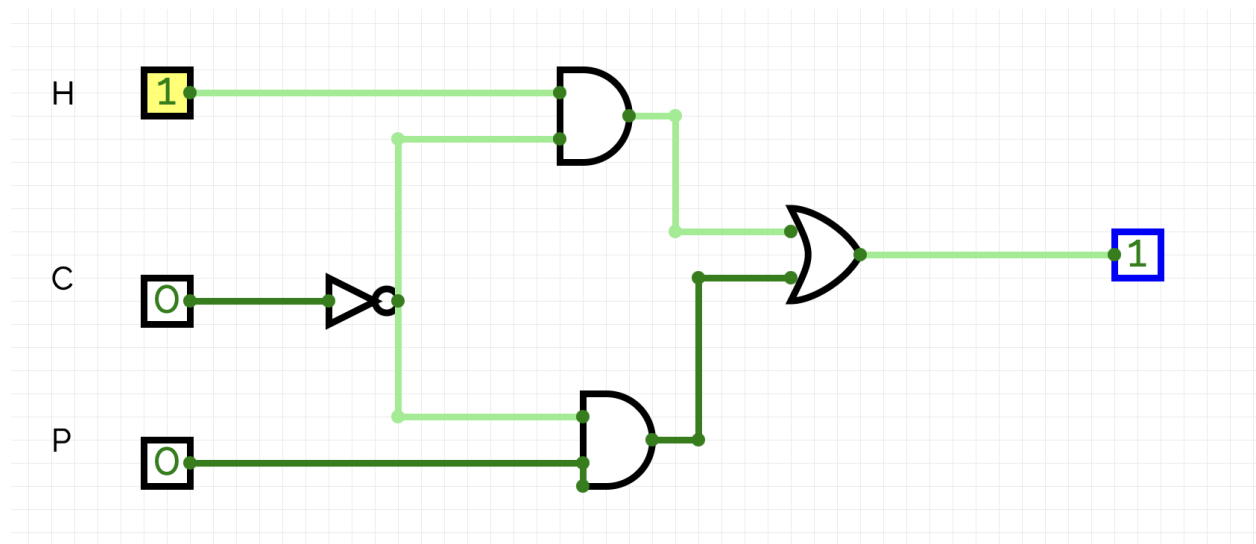
The reason that the Output is False is because there is no individual at the door and the button is not being pressed. While the door is unlocked it has the AND gate which means either H or P has to be True for the output to be True.

When $P = 0$, $H = 0$, $C = 1$, the expected is 0:



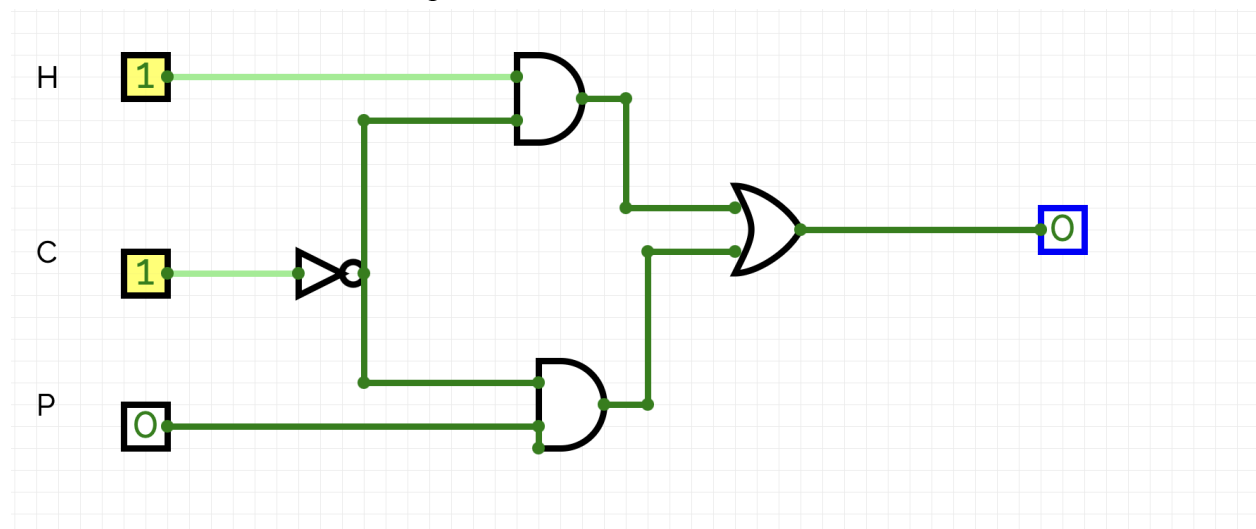
The reason that the Output is false is because there is no individual at the door and the button is not being pressed. The door is also locked so it wouldn't matter if there was an individual standing or the button is being pressed. The output will always be False until the door is unlocked and there is an individual standing at the door or the button is being pressed.

When $P = 0$, $H = 1$, $C = 0$, the expected is 1:



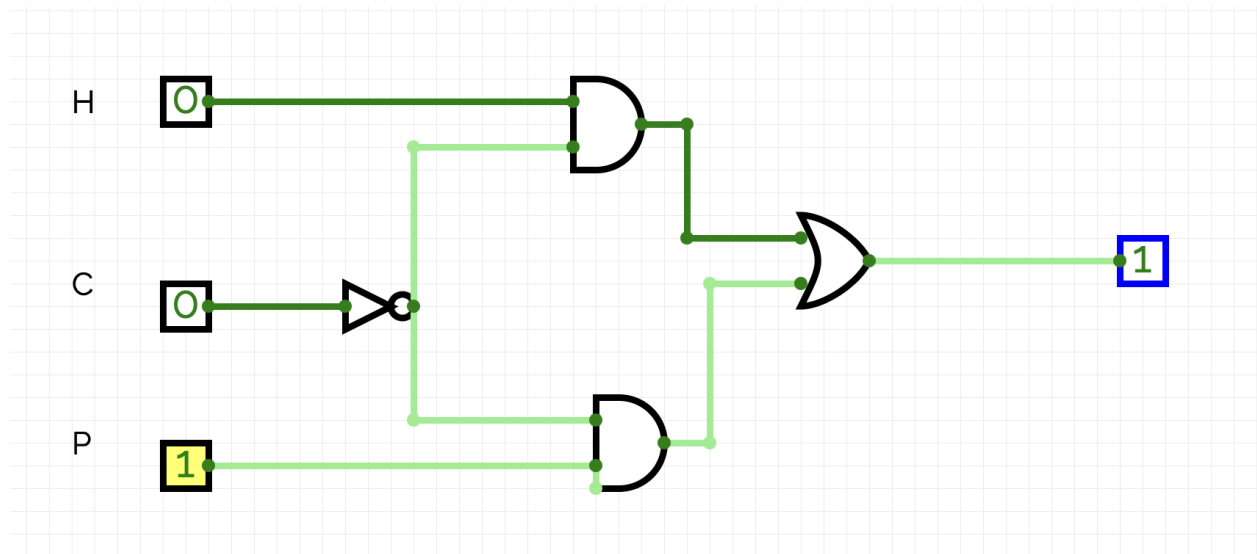
The reason that the Output is true is because the button is being pressed and the door is unlocked which would yield to it being true because the AND Gate is getting a true input. Which would make the inputs being received by the OR Gate being True and False, since it's True OR False the OR gate would receive the statement being True.

When $P = 0$, $H = 1$, $C = 1$, the expected is 0:



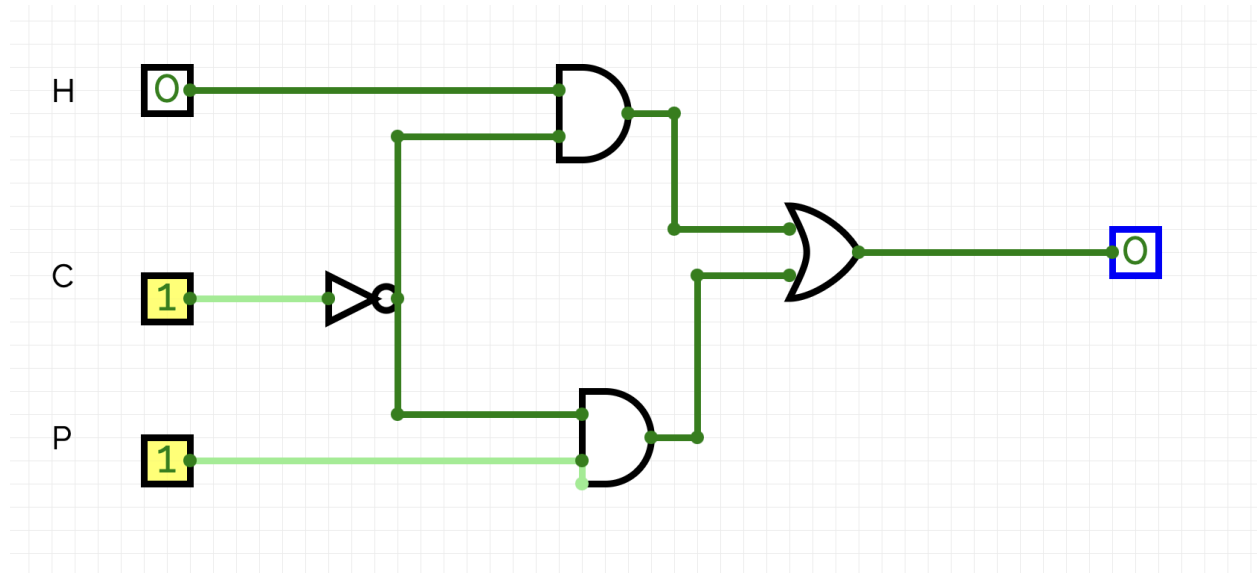
The reason that the output is False is because the Door is closed but the button is still being pressed which would lead the AND Gate to receiving an False input which would then go into the OR gate which is getting 2 false inputs. Meaning that the final output being seen is False

When $P = 1$, $H = 0$, $C = 0$, the expected is 1:



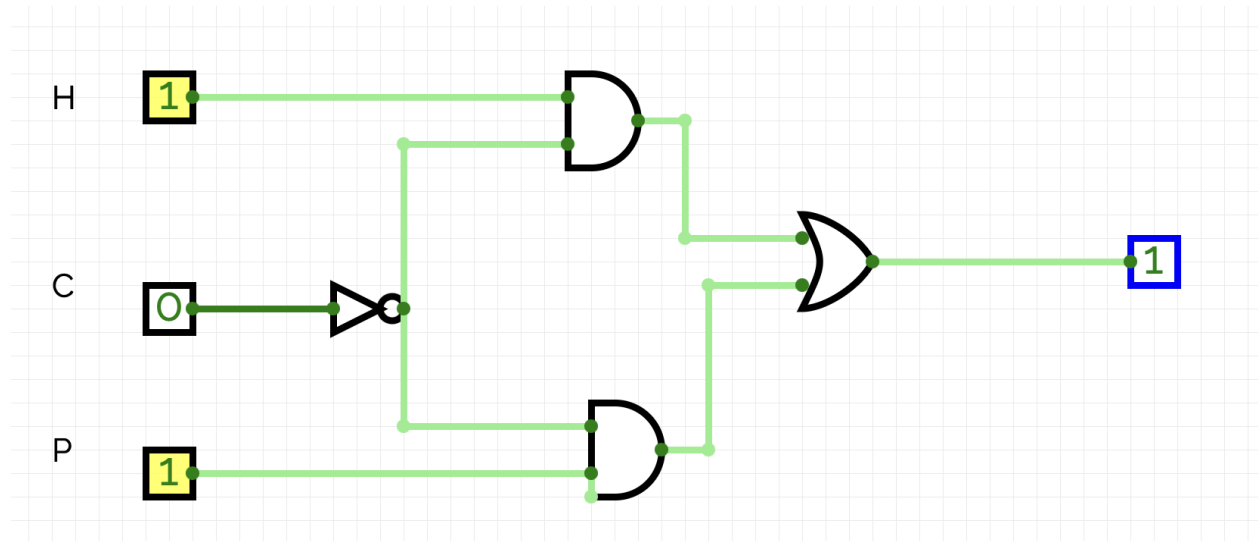
The reason that the output is True is because a person is being detected and the door is unlocked, which goes to the AND gate as true which then goes to the OR gate and is being received as true at the output.

When $P = 1$, $H = 0$, $C = 1$, the expected is 0:



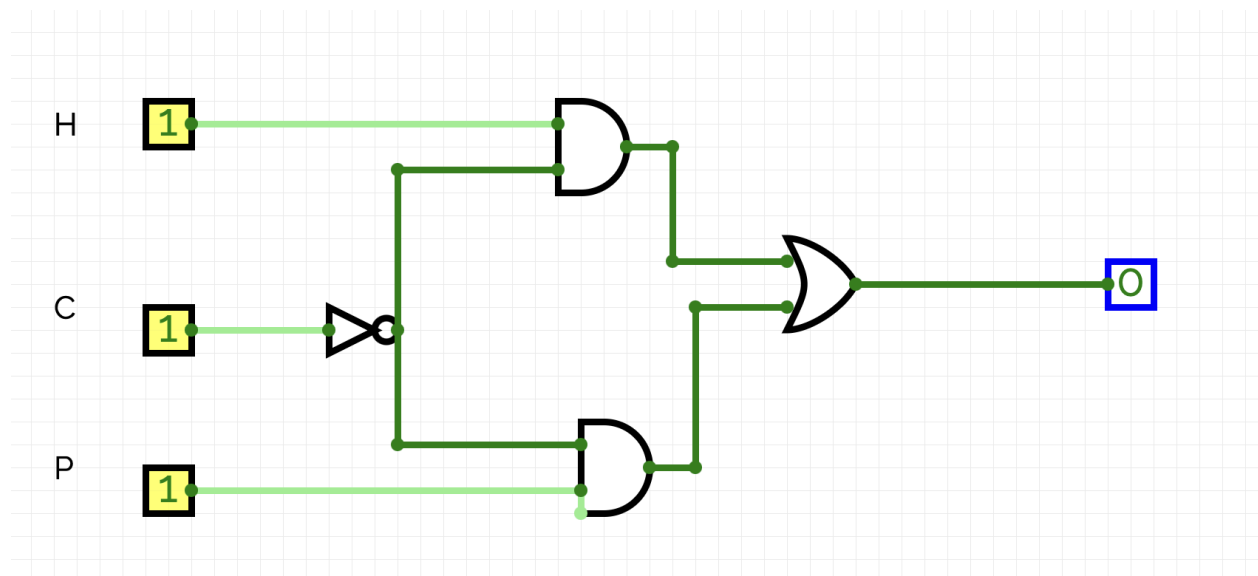
The output is False because the person is being detected but the door is closed which means that the AND gate is receiving False which then goes to the OR gate which is receiving False for both inputs which would go to the binary output.

When $P = 1$, $H = 1$, $C = 0$, the expected is 1:



The output is True because the button is being pressed and the person is being detected, while the door is unlocked, which means both AND gates are receiving a True statement which would then go onto the OR gate which outputs the statement as being True.

When $P = 1$, $H = 1$, $C = 1$, the expected is 0:



The output is False because the button is being pressed and the person is being detected but the door is closed. Which means when the statement goes to both the AND gates its being received as False which would make the options for the OR gate False and False which would lead the output to being False.